Ex. No: 11 Memory Allocation Methods

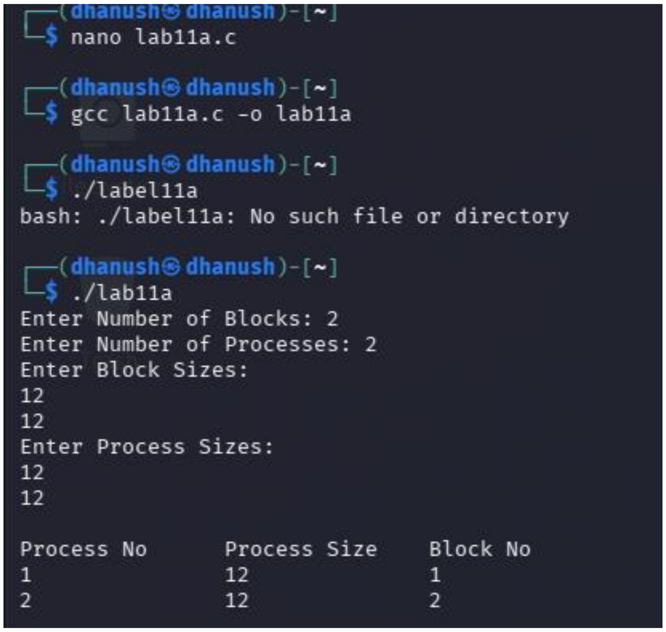
PROGRAM A: FIRST FIT

C PROGRAM

```
#include <stdio.h>
int main()
{
int blockSize[20], processSize[20];
int allocation[20];
int nb, np, i, j;
printf("Enter Number of Blocks: ");
scanf("%d",&nb);
printf("Enter Number of Processes: ");
scanf("%d",&np);
printf("Enter Block Sizes:\n");
for(i=0;i<nb;i++)
scanf("%d",&blockSize[i]);
printf("Enter Process Sizes:\n");
for(i=0;i<np;i++)
scanf("%d",&processSize[i]);
for(i=0;i<np;i++)
allocation[i]=-1;
for(i=0;i<np;i++)
{
for(j=0;j<nb;j++)
```

```
{
if(blockSize[j] >= processSize[i])
{
allocation[i]=j;
blockSize[j]-=processSize[i];
break;}
}
}
printf("\nProcess No\tProcess Size\tBlock No\n");
for(i=0;i<np;i++)
{
printf("%d\t\t%d\t\t",i+1,processSize[i]);
if(allocation[i]!=-1)
printf("%d\n",allocation[i]+1);
else
printf("Not Allocated\n");}
return 0;}
```

Output:



```
┌──(dhanush㉿dhanush)-[~]
└─$ nano lab11a.c

┌──(dhanush㉿dhanush)-[~]
└─$ gcc lab11a.c -o lab11a

┌──(dhanush㉿dhanush)-[~]
└─$ ./label11a
bash: ./label11a: No such file or directory

┌──(dhanush㉿dhanush)-[~]
└─$ ./lab11a
Enter Number of Blocks: 2
Enter Number of Processes: 2
Enter Block Sizes:
12
12
Enter Process Sizes:
12
12

Process No      Process Size    Block No
1               12              1
2               12              2
```

SHELL SCRIPT

#!/bin/bash

echo "First Fit Memory Allocation Demonstration"

blocks=(100 500 200 300 600)

processes=(212 417 112 426)

echo "Memory Blocks: ${blocks[@]}"

echo "Processes: ${processes[@]}"

echo "Allocation Performed Using First Fit"

Output:

```
┌──(dhanush㉿dhanush)-[~]
└─$ nano lab11a.sh

┌──(dhanush㉿dhanush)-[~]
└─$ chmod +x lab11a.sh

┌──(dhanush㉿dhanush)-[~]
└─$ bash lab11a.sh
First Fit Memory Allocation
Memory Blocks: 100 500 200 300 600
Processes: 212 417 112 426

Process Process Size    Block No
P1      212             B2
P2      417             B5
P3      112             B2
P4      426             Not Allocated
```

PROGRAM B: BEST FIT

C PROGRAM

```
#include <stdio.h>

int main()

{

int blockSize[20], processSize[20];

int allocation[20];

int nb,np,i,j,bestIdx;

printf("Enter Number of Blocks: ");

scanf("%d",&nb);

printf("Enter Number of Processes: ");

scanf("%d",&np);

printf("Enter Block Sizes:\n");

for(i=0;i<nb;i++)

scanf("%d",&blockSize[i]);

printf("Enter Process Sizes:\n");
```

```
for(i=0;i<np;i++)

scanf("%d",&processSize[i]);

for(i=0;i<np;i++)

allocation[i]=-1;

for(i=0;i<np;i++)

{

bestIdx=-1;

for(j=0;j<nb;j++)

{

if(blockSize[j] >= processSize[i])

{

if(bestIdx==-1 || blockSize[j] < blockSize[bestIdx])

bestIdx=j;

}

}

if(bestIdx!=-1)

{

allocation[i]=bestIdx;

blockSize[bestIdx]-=processSize[i];

}

}

printf("\nProcess No\tProcess Size\tBlock No\n");

for(i=0;i<np;i++)

{

printf("%d\t\t%d\t\t",i+1,processSize[i]);

if(allocation[i]!=-1)
```

printf("%d\n",allocation[i]+1);

else

printf("Not Allocated\n");

}

return 0;

}

Output:

```
┌──(dhanush㉿dhanush)-[~]
└─$ nano lab11b.c

┌──(dhanush㉿dhanush)-[~]
└─$ gcc lab11b.c -o lab11b

┌──(dhanush㉿dhanush)-[~]
└─$ ./lab11b
Enter Number of Blocks: 2
Enter Number of Processes: 2
Enter Block Sizes:
1
2
Enter Process Sizes:
1
2

Process No      Process Size    Block No
1               1               1
2               2               2
```

SHELL SCRIPT

#!/bin/bash

echo "Best Fit Memory Allocation Demonstration"

blocks=(100 500 200 300 600)

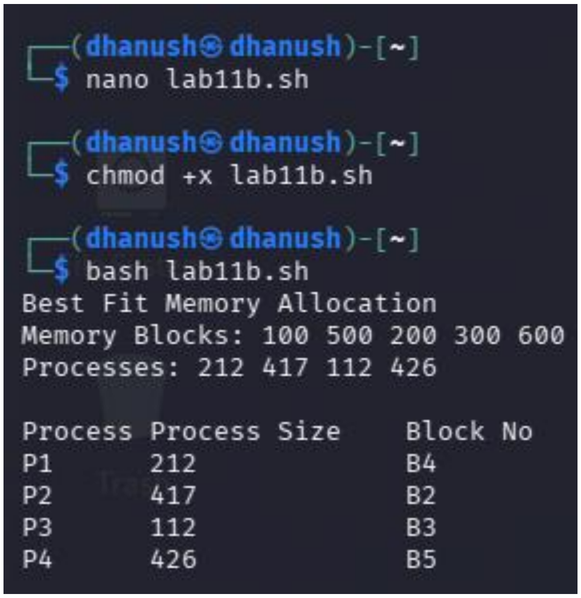
processes=(212 417 112 426)

echo "Memory Blocks: ${blocks[@]}"

echo "Processes: ${processes[@]}"

echo "Allocation Performed Using Best Fit"

Output:



```
┌──(dhanush㉿dhanush)-[~]
└─$ nano lab11b.sh

┌──(dhanush㉿dhanush)-[~]
└─$ chmod +x lab11b.sh

┌──(dhanush㉿dhanush)-[~]
└─$ bash lab11b.sh
Best Fit Memory Allocation
Memory Blocks: 100 500 200 300 600
Processes: 212 417 112 426

Process Process Size    Block No
P1      212             B4
P2      417             B2
P3      112             B3
P4      426             B5
```

PROGRAM C: WORST FIT

C PROGRAM

```
#include <stdio.h>

int main()

{

int blockSize[20], processSize[20];

int allocation[20];

int nb,np,i,j,worstIdx;

printf("Enter Number of Blocks: ");

scanf("%d",&nb);

printf("Enter Number of Processes: ");
```

```
scanf("%d",&np);
printf("Enter Block Sizes:\n");
for(i=0;i<nb;i++)
scanf("%d",&blockSize[i]);
printf("Enter Process Sizes:\n");
for(i=0;i<np;i++)
scanf("%d",&processSize[i]);
for(i=0;i<np;i++)
allocation[i]=-1;
for(i=0;i<np;i++)
{
worstIdx=-1;
for(j=0;j<nb;j++)
{
if(blockSize[j] >= processSize[i])
{
if(worstIdx==-1 || blockSize[j] > blockSize[worstIdx])
worstIdx=j;
}
}
if(worstIdx!=-1)
{
allocation[i]=worstIdx;
```

```
blockSize[worstIdx]-=processSize[i];
}
}
printf("\nProcess No\tProcess Size\tBlock No\n");
for(i=0;i<np;i++)
{
printf("%d\t\t%d\t\t",i+1,processSize[i]);
if(allocation[i]!=-1)
printf("%d\n",allocation[i]+1);
else
printf("Not Allocated\n");
}
return 0;
}
```

Output:

```
┌──(dhanush㉿dhanush)-[~]
└─$ nano lab11c.c

┌──(dhanush㉿dhanush)-[~]
└─$ gcc lab11c.c -o lab11c

┌──(dhanush㉿dhanush)-[~]
└─$ ./lab11
bash: ./lab11: No such file or directory

┌──(dhanush㉿dhanush)-[~]
└─$ ./lab11c
Enter Number of Blocks: 2
Enter Number of Processes: 2
Enter Block Sizes:
12
12
Enter Process Sizes:
12
12

Process No      Process Size    Block No
1               12              1
2               12              2
```

SHELL SCRIPT

```
#!/bin/bash

echo "Worst Fit Memory Allocation Demonstration"

blocks=(100 500 200 300 600)

processes=(212 417 112 426)

echo "Memory Blocks: ${blocks[@]}"

echo "Processes: ${processes[@]}"

echo "Allocation Performed Using Worst Fit"
```

Output:

```
┌──(dhanush㉿dhanush)-[~]
└─$ nano lab11.sh

┌──(dhanush㉿dhanush)-[~]
└─$ chmod +x lab11.sh

┌──(dhanush㉿dhanush)-[~]
└─$ bash lab11.sh
Worst Fit Memory Allocation
Memory Blocks: 100 500 200 300 600
Processes: 212 417 112 426

Process Process Size    Block No
P1      212             B5
P2      417             B2
P3      112             B5
P4      426             Not Allocated
```